

Computer Organization & Design + Computer Networking

Complete Interview Q&A Guide — Basic to Advanced

Covers every concept from **Computer Architecture, Digital Logic, Memory Systems, CPU Design, Networking Protocols, OSI/TCP-IP Models, Routing, Security**, and more.

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PART A — Computer Organization & Design

1. Digital Logic & Number Systems

Q1. What are the different number systems used in computing?

A: Computers internally process binary (base-2), but humans and systems communicate in various bases:

System	Base	Digits	Example
Binary	2	0, 1	1011
Octal	8	0–7	13
Decimal	10	0–9	11
Hexadecimal	16	0–9, A–F	0xB

All four represent the same value: decimal **11**.

Conversions:

- **Decimal** → **Binary**: Repeated division by 2; remainders in reverse.
 - **Binary** → **Hex**: Group bits in sets of 4 from right.
 - **Binary** → **Octal**: Group bits in sets of 3 from right.
-

Q2. How is negative numbers represented in binary?

A: Three main representations:

1. **Sign-Magnitude**: MSB = sign bit (0 = positive, 1 = negative). Problem: two zeros (+0 and -0).
2. **One's Complement**: Invert all bits. Still has two zeros.
3. **Two's Complement** (*most widely used*): Invert all bits, then add 1. Only one zero, arithmetic is simpler.

```
+5 = 0000 0101
-5 in Two's Complement:
  Invert: 1111 1010
```

Add 1: 1111 1011 ← This is -5

Two's complement for n-bit: range = $-2^{(n-1)}$ to $2^{(n-1)} - 1$. For 8-bit: -128 to +127.

Q3. What is overflow in binary arithmetic?

A: Overflow occurs when the result of an arithmetic operation exceeds the range representable in the given number of bits.

Signed overflow detection: Overflow has occurred if:

- Two positive numbers are added and the result is negative.
- Two negative numbers are added and the result is positive.

In two's complement: overflow if carry into MSB $\neq$ carry out of MSB.

Q4. What is the difference between signed and unsigned integers?

A:

- **Unsigned:** All bits represent magnitude. Range: 0 to $2^n - 1$. (e.g., 8-bit: 0–255)
- **Signed** (two's complement): MSB is sign bit. Range: $-2^{(n-1)}$ to $2^{(n-1)} - 1$. (e.g., 8-bit: -128–127)

Same bit pattern 1111 1111 means 255 (unsigned) or -1 (signed 8-bit).

Q5. How is floating-point represented in computers (IEEE 754)?

A: IEEE 754 is the standard for floating-point arithmetic.

32-bit single precision:

| Sign (1 bit) | Exponent (8 bits) | Mantissa/Fraction (23 bits) |

- **Sign:** 0 = positive, 1 = negative.
- **Exponent:** Biased by 127 (stored value = actual exponent + 127).
- **Mantissa:** Fractional part of normalized significand (1.fraction).

Value = $(-1)^{\text{sign}} \times 1.\text{mantissa} \times 2^{(\text{exponent} - 127)}$

Special values: 0 (all zeros), Infinity (exponent all 1s, mantissa 0), NaN (exponent all 1s, mantissa $\neq$ 0).

Q6. What is BCD (Binary Coded Decimal)?

A: BCD encodes each decimal digit as its 4-bit binary equivalent, making decimal arithmetic and human-readable output simpler.

Decimal: 2 5 7
BCD: 0010 0101 0111

Used in: financial systems, clocks, calculators where exact decimal representation matters and floating-point errors are unacceptable.

2. Boolean Algebra & Logic Gates

Q7. What are the basic logic gates?

A:

Gate	Symbol	Operation	Output
AND	$A \cdot B$	Both inputs 1	1 only if A=1 AND B=1
OR	$A + B$	Either input 1	1 if A=1 OR B=1
NOT	$\bar{A}$	Invert	Opposite of input
NAND	$\neg(A \cdot B)$	NOT AND	0 only if A=1 AND B=1
NOR	$\neg(A + B)$	NOT OR	1 only if both 0
XOR	$A \oplus B$	Exclusive OR	1 if inputs differ
XNOR	$\neg(A \oplus B)$	Exclusive NOR	1 if inputs same

NAND and NOR are universal gates — any Boolean function can be built from NAND-only or NOR-only circuits.

Q8. What are the key theorems of Boolean Algebra?

A:

Theorem	Expression
Identity	$A + 0 = A; A \cdot 1 = A$
Null	$A + 1 = 1; A \cdot 0 = 0$
Idempotent	$A + A = A; A \cdot A = A$
Complement	$A + \bar{A} = 1; A \cdot \bar{A} = 0$
Double Negation	$\neg(\neg A) = A$
Commutative	$A + B = B + A; A \cdot B = B \cdot A$
Associative	$(A + B) + C = A + (B + C)$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$
De Morgan's	$\neg(A \cdot B) = \bar{A} + \bar{B}; \neg(A + B) = \bar{A} \cdot \bar{B}$
Absorption	$A + A \cdot B = A; A \cdot (A + B) = A$

De Morgan's Law is critical for converting between NAND/NOR and AND/OR/NOT implementations.

Q9. What is a Karnaugh Map (K-Map)?

A: A K-Map is a visual method for simplifying Boolean expressions by grouping adjacent 1s (for SOP) or 0s (for POS) in a grid.

Rules:

- Groups must be powers of 2: 1, 2, 4, 8, 16.
- Groups can wrap around edges (toroidal adjacency).
- Larger groups → fewer terms → simpler expression.
- Every 1 must be covered; groups should be as large as possible.

K-Map for $F(A, B, C, D)$:

AB	CD			
	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	0	1	1	0
10	0	0	1	0

Grouping the column $CD=11$ gives: $F = CD$ (simplified from a complex sum of products).

Q10. What is Sum of Products (SOP) and Product of Sums (POS)?

A:

- **SOP (Sum of Products):** OR of AND terms (minterms). Derived from rows where output = 1.
 - Example: $F = AB + B\bar{C} + \bar{A}C$
- **POS (Product of Sums):** AND of OR terms (maxterms). Derived from rows where output = 0.
 - Example: $F = (A+B)(\bar{B}+C)(\bar{A}+\bar{C})$

Both are canonical forms; K-maps can simplify either.

3. Combinational Circuits

Q11. What is a multiplexer (MUX)?

A: A MUX is a **data selector** — it selects one of N inputs and routes it to the output based on S select lines (where $N = 2^S$).

A 4-to-1 MUX has 4 inputs (D_0 – D_3), 2 select lines (S_1, S_0), and 1 output.

S1	S0	Output
0	0	D0
0	1	D1
1	0	D2
1	1	D3

MUX can implement any Boolean function — it's a universal logic element.

Q12. What is a demultiplexer (DEMUX)?

A: A DEMUX routes a single input to one of N outputs based on select lines. Opposite of MUX.

A 1-to-4 DEMUX: 1 input, 2 select lines, 4 outputs. Only selected output carries the input; others are 0.

Q13. What is an encoder and a decoder?

A:

- **Encoder:** Converts 2^N input lines to N-bit binary code. Example: 8-to-3 encoder compresses 8 active-low lines to 3-bit binary.
- **Decoder:** Converts N-bit binary code to one of 2^N output lines. Example: 3-to-8 decoder activates exactly one of 8 outputs.
- **Priority Encoder:** If multiple inputs active, gives output for highest-priority input.

Decoders are used in memory addressing — selecting specific memory chips or registers.

Q14. What is a half adder and full adder?

A:

- **Half Adder:** Adds two 1-bit inputs. Produces Sum = $A \oplus B$, Carry = $A \cdot B$. Cannot handle carry-in.
- **Full Adder:** Adds two bits + carry-in. Sum = $A \oplus B \oplus C_{in}$, Cout = $A \cdot B + C_{in} \cdot (A \oplus B)$.

Ripple Carry Adder: Chain of full adders. Simple but slow — carry propagates bit-by-bit ($O(n)$ delay).

Carry Lookahead Adder (CLA): Pre-computes carries in parallel using generate ($G=A \cdot B$) and propagate ($P=A \oplus B$) signals. Much faster — $O(\log n)$ delay.

Q15. What is a comparator circuit?

A: A digital circuit that compares two binary numbers A and B and produces outputs indicating:

- $A > B$
- $A = B$
- $A < B$

A 1-bit comparator: $A=B$ when $A \text{ XNOR } B = 1$. For multi-bit: compare MSB first; if equal, compare next bit, and so on. Iterative comparators chain 1-bit comparators.

4. Sequential Circuits

Q16. What is the difference between combinational and sequential circuits?

A:

Aspect	Combinational	Sequential
Memory	None	Has memory (state)
Output depends on	Current inputs only	Current inputs + current state
Clock	Not needed	Usually needed
Examples	Adder, MUX, Decoder	Flip-flop, Counter, Register

Sequential circuits have **feedback** — output is fed back as input.

Q17. What are flip-flops? Explain types.

A: A flip-flop is a 1-bit memory cell — the fundamental building block of sequential logic.

SR Flip-Flop (Set-Reset):

- $S=1, R=0 \rightarrow Q=1$ (Set)
- $S=0, R=1 \rightarrow Q=0$ (Reset)
- $S=0, R=0 \rightarrow Q$ unchanged (Hold)
- $S=1, R=1 \rightarrow$ Forbidden (invalid state)

D Flip-Flop (Data):

- Q follows D on the clock edge. Most commonly used.
- Eliminates the forbidden state of SR.

JK Flip-Flop:

- Improvement of SR: $J=1, K=1 \rightarrow$ Toggle (no forbidden state).
- Most versatile flip-flop.

T Flip-Flop (Toggle):

- $T=1 \rightarrow$ Toggle; $T=0 \rightarrow$ Hold.
 - Used in counters.
-

Q18. What is the difference between level-triggered and edge-triggered flip-flops?

A:

- **Level-triggered (Latch):** Output changes whenever the enable/clock signal is high (or low). Transparent — input change immediately affects output during active level.
- **Edge-triggered:** Output changes only at the rising edge (positive edge) or falling edge (negative edge) of the clock. More predictable and stable.

Most modern digital systems use **positive edge-triggered** D flip-flops to avoid timing hazards.

Q19. What are registers and counters?

A:

- **Register:** A group of flip-flops storing multiple bits of data. An 8-bit register holds one byte. Types: parallel load register, shift register.
 - **Shift Register:** Bits shift left or right on each clock cycle. Types: SISO, SIPO, PISO, PIPO. Used for serial-to-parallel conversion, data transmission.
 - **Counter:** Sequential circuit that counts clock pulses.
 - **Ripple (Asynchronous) Counter:** Each FF clocked by previous FF output. Simple but has propagation delay.
 - **Synchronous Counter:** All FFs clocked simultaneously. Faster and more reliable.
 - **Up/Down Counter, Modulo-N Counter** (counts 0 to N-1).
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Q20. What are hazards in digital circuits?

A: Hazards are **unwanted transient glitches** in circuit output due to unequal propagation delays through different paths.

- **Static-1 Hazard:** Output should remain 1 but briefly glitches to 0.
- **Static-0 Hazard:** Output should remain 0 but briefly glitches to 1.
- **Dynamic Hazard:** Output changes more than once when it should change only once.

Elimination: Add **redundant gates** (consensus terms) to K-Map groups to cover transitions between adjacent cells. Synchronous design reduces hazard impact by only sampling outputs at clock edges.

5. Computer Organization Basics

Q21. What are the basic components of a computer system?

A: A computer consists of five fundamental components:

1. **CPU (Central Processing Unit):** Fetches, decodes, and executes instructions. Contains ALU, CU, and registers.

2. **Memory (RAM):** Stores instructions and data currently in use. Volatile.
3. **Storage (HDD/SSD):** Persistent storage for programs and files. Non-volatile.
4. **Input Devices:** Keyboard, mouse, scanner.
5. **Output Devices:** Monitor, printer, speakers.

The **Von Neumann architecture** connects these via a shared bus (address, data, control).

Q22. What is Von Neumann Architecture?

A: Proposed by John Von Neumann in 1945. Key characteristics:

- A single memory stores both **instructions and data** (stored-program concept).
- Components communicate via a shared **system bus**.
- Sequential instruction execution (fetch → decode → execute cycle).

Von Neumann Bottleneck: CPU is much faster than memory; the shared bus between CPU and memory becomes the performance bottleneck.

Harvard Architecture (alternative): Separate memories and buses for instructions and data. Used in embedded systems (AVR, PIC) and cache designs.

Q23. What is the difference between RISC and CISC architectures?

A:

Aspect	RISC	CISC
Full form	Reduced Instruction Set Computer	Complex Instruction Set Computer
Instructions	Simple, fixed-length	Complex, variable-length
Execution	1 clock cycle per instruction (ideal)	Multiple clock cycles
Addressing modes	Few	Many
Registers	Many	Fewer
Memory access	Load/Store only	Any instruction can access memory
Pipeline-friendly	Yes	Less so
Examples	ARM, MIPS, RISC-V	x86, x86-64

Modern CISC (like x86) uses a micro-op layer internally that translates CISC instructions to RISC-like micro-ops for efficient pipelining.

Q24. What is the system bus? What are its types?

A: The system bus is the communication pathway connecting CPU, memory, and I/O. It has three components:

- **Address Bus:** Unidirectional (CPU → memory/I/O). Carries memory addresses. Width determines addressable memory (32-bit = 4 GB).
 - **Data Bus:** Bidirectional. Carries actual data. Width = word size (8, 16, 32, 64 bits).
 - **Control Bus:** Bidirectional. Carries control signals: read/write, interrupt, clock, reset, etc.
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Q25. What is word size in a computer?

A: Word size is the natural unit of data processed by a CPU — typically equal to the width of the data bus and general-purpose registers.

- 8-bit (early microprocessors)
- 16-bit (Intel 8086)
- 32-bit (x86 — IA-32)
- 64-bit (x86-64, ARM64)

Word size affects: maximum addressable memory, integer precision, register width, pointer size.

6. CPU Architecture & Design

Q26. What are the internal components of a CPU?

A:

1. **ALU (Arithmetic Logic Unit):** Performs arithmetic (+, -, *, /) and logical (AND, OR, NOT, XOR, compare, shift) operations.
 2. **Control Unit (CU):** Fetches instructions from memory, decodes them, and generates control signals to coordinate all components.
 3. **Registers:** Ultra-fast, small storage inside the CPU.
 4. **Program Counter (PC):** Holds address of the next instruction.
 5. **Instruction Register (IR):** Holds the currently executing instruction.
 6. **Memory Address Register (MAR):** Holds address to be accessed in memory.
 7. **Memory Data Register (MDR/MBR):** Holds data read from or written to memory.
 8. **Status/Flags Register:** Holds condition codes (Zero, Carry, Overflow, Sign, Parity).
-

Q27. Explain the Fetch-Decode-Execute cycle.

A: The instruction cycle is the fundamental operation of a CPU:

1. **Fetch:** PC → MAR → Memory reads instruction → MDR → IR. PC incremented.
2. **Decode:** CU interprets the opcode in IR. Identifies instruction type, operands, addressing mode.
3. **Execute:** ALU or other units perform the operation. Results written to registers or memory.
4. **Write-back:** Results stored in destination register or memory (sometimes merged with execute).

This cycle repeats continuously until a HALT instruction or interrupt.

Q28. What are the types of CPU registers?

A:

Type	Purpose
General-Purpose Registers (GPR)	Hold data/addresses for computation (EAX, EBX in x86)
Program Counter (PC)	Address of next instruction
Stack Pointer (SP)	Top of current stack frame
Base/Frame Pointer (BP)	Base of current stack frame
Index Registers (SI, DI)	Array indexing, string operations
Flags/Status Register	Condition codes after ALU ops
Instruction Register (IR)	Currently executing instruction
Segment Registers	Memory segment base addresses (x86 legacy)
Floating-Point Registers	FP arithmetic (XMM, YMM in x86)

Q29. What are addressing modes?

A: Addressing modes specify how an instruction identifies its operand(s):

Mode	Description	Example
Immediate	Operand is in the instruction itself	MOV AX, 5 (AX = 5)
Register	Operand is in a register	MOV AX, BX
Direct	Instruction contains memory address	MOV AX, [1000H]
Indirect	Register holds the address	MOV AX, [BX]
Register Indirect	Register holds pointer	MOV AX, [BX]
Indexed	Base address + index register	MOV AX, [BX + SI]
Based Indexed	Base + index + displacement	MOV AX, [BX + SI + 4]
Relative	PC + offset (branches/jumps)	JMP +10

Q30. What is interrupt handling? Types of interrupts?

A: An interrupt is a signal that temporarily halts the CPU's current task to handle an urgent event.

Process:

1. Interrupt signal received.
2. CPU completes current instruction.
3. Current state (PC, registers) saved to stack.
4. ISR (Interrupt Service Routine) address loaded from Interrupt Vector Table.
5. ISR executes.
6. State restored, normal execution resumes.

Types:

- **Hardware Interrupt:** From external devices (keyboard press, timer, disk I/O complete). Asynchronous.
- **Software Interrupt (Trap):** Triggered by instructions (INT 21H in x86). Used for system calls.
- **Exception:** Internal CPU error (divide by zero, page fault, invalid opcode). Synchronous.
- **Non-Maskable Interrupt (NMI):** Cannot be disabled. Used for critical hardware failures.
- **Maskable Interrupt:** Can be enabled/disabled by the CPU (via interrupt flag).

7. Instruction Set Architecture (ISA)

Q31. What is an ISA (Instruction Set Architecture)?

A: ISA is the abstract interface between software and hardware — it defines what the CPU can do from a programmer's/compiler's perspective:

- Instruction formats and opcodes
- Data types supported
- Registers available
- Addressing modes
- Memory model (byte ordering, alignment)
- Interrupt and exception mechanisms

Examples: x86, x86-64, ARM, MIPS, RISC-V, SPARC.

Q32. What is the difference between big-endian and little-endian?

A: Endianness defines how multi-byte values are stored in memory:

- **Big-Endian:** Most significant byte at the lowest address.
 - 0x12345678 stored as: 12 34 56 78
 - Used by: MIPS, SPARC, network protocols (network byte order).
- **Little-Endian:** Least significant byte at the lowest address.
 - 0x12345678 stored as: 78 56 34 12
 - Used by: x86, x86-64, ARM (configurable).

When transmitting data over a network (which uses big-endian), little-endian systems must convert using `htonl()`/`ntohl()` functions.

Q33. What are the types of CPU instructions?

A:

1. **Data Transfer:** MOV, LOAD, STORE, PUSH, POP, XCHG
 2. **Arithmetic:** ADD, SUB, MUL, DIV, INC, DEC
 3. **Logical:** AND, OR, NOT, XOR, TEST
 4. **Shift/Rotate:** SHL, SHR, SAL, SAR, ROL, ROR
 5. **Branch/Jump:** JMP, JE, JNE, JG, JL, CALL, RET
 6. **Compare:** CMP (sets flags without modifying registers)
 7. **I/O:** IN, OUT (x86)
 8. **System:** INT (system call), HLT (halt), NOP (no operation), CPUID
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8. Pipelining

Q34. What is CPU pipelining?

A: Pipelining is a technique where multiple instructions are overlapped in execution, similar to an assembly line. While one instruction is being executed, the next is being decoded, and the one after is being fetched.

Classic 5-Stage RISC Pipeline (MIPS):

```
Stage 1: IF – Instruction Fetch
Stage 2: ID – Instruction Decode / Register Read
Stage 3: EX – Execute (ALU)
Stage 4: MEM – Memory Access
Stage 5: WB – Write Back to Register
```

Ideal speedup: With k stages, k instructions complete simultaneously $\rightarrow k\times$ throughput (but

latency of single instruction unchanged).

Q35. What are pipeline hazards?

A: Hazards are situations that prevent the next instruction from executing in the immediately following clock cycle.

1. Structural Hazard: Two instructions need the same hardware resource simultaneously.

- Solution: Stall pipeline, or add more hardware (e.g., separate instruction and data caches).

2. Data Hazard: An instruction needs data that hasn't been computed yet.

- **RAW (Read After Write):** True dependency — most common.
- **WAR (Write After Read):** Anti-dependency.
- **WAW (Write After Write):** Output dependency.
- Solutions: **Stalling (bubbles)**, **Forwarding/Bypassing** (pass result directly from EX to next instruction's EX stage without waiting for WB), **Out-of-Order Execution**.

3. Control Hazard (Branch Hazard): Branch instruction — CPU doesn't know the next instruction until the branch is resolved.

- Solutions: **Pipeline flush**, **Branch prediction** (static/dynamic), **Delayed branching** (RISC — execute one instruction after branch regardless).
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Q36. What is branch prediction?

A: Branch prediction guesses whether a branch will be taken or not to keep the pipeline full.

- **Static Prediction:** Always predict taken (loops are usually taken), or use compiler hints.
- **Dynamic Prediction** — hardware-based:
 - **1-bit predictor:** Single bit remembers last outcome.
 - **2-bit saturating counter:** Changes prediction only after two consecutive mispredictions. More stable for loops.
 - **Correlating / Local / Global History:** Uses history tables to exploit patterns.
 - **Tournament Predictor:** Combines multiple predictors, selects the better one per branch.

Modern CPUs (Intel, AMD) achieve ~95%+ prediction accuracy with advanced predictors.

Q37. What is out-of-order execution?

A: A technique where the CPU executes instructions in a different order than the program specifies to avoid stalls from data hazards, as long as the logical result is the same.

Key concepts:

- **Register Renaming:** Eliminates WAR and WAW hazards by mapping logical registers to a larger set of physical registers.
- **Reservation Stations:** Instructions wait for their operands; execute as soon as operands are ready.
- **Reorder Buffer (ROB):** Ensures results are committed in program order even if executed out of order.
- **Tomasulo's Algorithm:** Classic OOO execution algorithm.

Q38. What is superscalar architecture?

A: A superscalar CPU can **fetch, decode, and execute multiple instructions per clock cycle** by having multiple execution units (ALUs, FPU, load/store units) operating in parallel.

- **ILP (Instruction-Level Parallelism):** Exploit parallelism within a single instruction stream.
- **Issue Width:** Number of instructions issued per cycle (2-wide, 4-wide, 6-wide).
- Modern Intel/AMD CPUs are 4–6-wide superscalar.

Contrast with **VLIW (Very Long Instruction Word)**: Compiler packs multiple operations into one wide instruction; hardware is simpler but compiler is more complex. Used in DSPs.

9. Memory Hierarchy & Cache

Q39. What is the memory hierarchy?

A: Memory hierarchy organizes storage by speed, cost, and capacity:

Fastest/Smallest/Most Expensive		
↓	CPU Registers	(< 1 ns, bytes)
↓	L1 Cache	(1–4 ns, 32–64 KB per core)
↓	L2 Cache	(5–10 ns, 256 KB–1 MB)
↓	L3 Cache	(20–40 ns, 8–64 MB, shared)
↓	Main Memory (DRAM)	(60–100 ns, 8–256 GB)
↓	SSD/NVMe	(100 µs, TB)
↓	HDD	(10 ms, TB)
↓	Network/Tape Storage	(seconds, PB)
Slowest/Largest/Cheapest		

Principle of Locality:

- **Temporal locality:** Recently accessed data is likely to be accessed again soon.
- **Spatial locality:** Data near recently accessed data is likely to be accessed soon.

Cache exploits both principles.

Q40. What is a cache? How does it work?

A: Cache is a small, fast memory between the CPU and main memory that stores frequently used data/instructions.

Cache Hit: Data found in cache. CPU reads from cache (fast). **Cache Miss:** Data not in cache. CPU fetches from memory, loads into cache (slow).

Hit Rate = Cache Hits / Total Accesses **Average Access Time** = Hit Rate × Cache Time + Miss Rate × Memory Time

Cache Organization:

- **Tag:** Identifies which memory block is stored.
 - **Data Block:** Actual cached data.
 - **Valid Bit:** Whether the entry is valid.
 - **Dirty Bit:** Whether the cached copy has been modified (write-back caches).
-

Q41. What are cache mapping techniques?

A:

1. Direct Mapped Cache: Each memory block maps to exactly one cache line.

- Fast lookup, simple hardware.
- **Conflict misses:** Two frequently used blocks map to same line, thrash each other.

2. Fully Associative Cache: A block can go in any cache line.

- Best miss rate, but requires searching all lines simultaneously (expensive hardware — CAM).

3. Set-Associative Cache: Compromise. Cache divided into sets; each memory block maps to one set, can go in any line within that set (N-way = N lines per set).

- 2-way, 4-way, 8-way are common.
- Better than direct mapped; simpler than fully associative.

Address decomposition (for direct mapped):

[Tag | Index | Block Offset]

Q42. What are cache replacement policies?

A: When a cache set is full and a new block must be loaded:

- **LRU (Least Recently Used):** Replace the block not used for the longest time. Best performance, moderate hardware cost.
- **LFU (Least Frequently Used):** Replace block with fewest accesses. Can retain stale blocks.
- **FIFO (First In, First Out):** Replace oldest block regardless of use.
- **Random:** Replace a randomly selected block. Simple; surprisingly competitive.

- **PLRU (Pseudo-LRU):** Approximation of LRU using fewer bits. Used in practice (e.g., x86 L1 cache).
-

Q43. What are cache write policies?

A:

Write Hit Policies:

- **Write-Through:** Simultaneously writes to cache and main memory. Memory always up to date. Simple but generates more bus traffic.
- **Write-Back (Copy-Back):** Writes only to cache; memory updated when block is evicted (dirty bit marks modified blocks). Faster, less traffic, but memory can be stale.

Write Miss Policies:

- **Write-Allocate:** On miss, load block into cache, then write. Usually paired with write-back.
 - **No-Write-Allocate:** Write directly to memory without loading block into cache. Usually paired with write-through.
-

Q44. What are the types of cache misses (3 C's)?

A:

1. **Compulsory (Cold) Misses:** First access to a block — unavoidable. Reduced by prefetching.
 2. **Capacity Misses:** Cache too small to hold all needed blocks. Reduced by larger cache.
 3. **Conflict Misses:** Multiple blocks compete for the same cache line (direct mapped / low-associativity). Reduced by higher associativity.
 4. **Coherence Misses (4th C):** In multiprocessors, a cache line is invalidated because another processor wrote to the same block. Reduced by better coherence protocols.
-

10. Virtual Memory

Q45. What is virtual memory?

A: Virtual memory is a memory management technique that gives each process the **illusion of having its own large, contiguous address space**, independent of physical RAM.

Benefits:

- Programs can be larger than physical memory.
- Memory isolation between processes (security).
- Simplifies programming (no need to manage physical addresses).

- Allows multiprogramming — multiple processes in memory simultaneously.

Virtual addresses are translated to physical addresses by the **MMU (Memory Management Unit)** using a **page table**.

Q46. What is paging?

A: Paging divides virtual memory into fixed-size blocks called **pages** and physical memory into same-size blocks called **frames**.

- Typical page size: 4 KB (also 2 MB, 1 GB huge pages).
- Page table maps virtual page numbers (VPN) to physical frame numbers (PFN).
- **Page Table Entry (PTE):** Contains PFN + valid bit + dirty bit + access bits + protection bits.

Address translation:

```
Virtual Address = [VPN | Page Offset]
Physical Address = [PFN | Page Offset]    (offset unchanged)
```

Q47. What is a TLB (Translation Lookaside Buffer)?

A: The TLB is a hardware **cache for page table entries** — it stores recently used VPN → PFN mappings to avoid accessing the full page table in memory on every instruction.

- **TLB Hit:** VPN found in TLB → immediate translation. Very fast.
- **TLB Miss:** Walk the page table in memory (hardware or software page table walker). Load result into TLB.
- **TLB Flush:** On context switch, TLB entries must be invalidated (unless ASID — Address Space IDs — are used to tag entries).

TLB hit rate typically > 99% due to spatial and temporal locality.

Q48. What is a page fault?

A: A page fault occurs when the CPU accesses a virtual page not currently mapped to a physical frame (valid bit = 0 in PTE).

Handling:

1. OS page fault handler invoked.
2. Check if access is valid (in process's address space). If not → segfault/access violation.
3. Find a free frame (or evict a page using replacement policy).
4. Load the required page from disk (swap space).
5. Update page table entry.
6. Restart the faulting instruction.

Page replacement algorithms: FIFO, LRU, Optimal (OPT — theoretical), Clock (approximates LRU), Working Set.

Q49. What is segmentation?

A: Segmentation divides the address space into variable-size logical segments (code, data, stack, heap), each with a base address and limit.

- **Logical address:** [Segment Number | Offset]
- Segment table maps segment numbers to (base, limit) pairs.
- Provides natural protection: separate segments for code (read-only), data, stack.

Problem: External fragmentation (gaps between segments). Many modern OSes use paging exclusively or **segmented paging** (Intel x86 used both).

11. Input/Output Systems

Q50. What are the I/O techniques in computer architecture?

A:

1. Programmed I/O (Polling):

- CPU continuously checks (polls) the I/O device status register.
- Simple but wasteful — CPU spins doing nothing useful. CPU utilization low.

2. Interrupt-Driven I/O:

- CPU initiates I/O, then continues other work.
- Device interrupts CPU when operation completes.
- Better CPU utilization for slow devices. Overhead for each interrupt.

3. DMA (Direct Memory Access):

- A **DMA controller** transfers data directly between I/O device and memory without CPU involvement.
 - CPU only involved at start and end of transfer.
 - Best for high-bandwidth transfers (disk, network).
 - DMA can steal memory bus cycles from CPU (**cycle stealing**).
-

Q51. What is memory-mapped I/O vs port-mapped I/O?

A:

- **Memory-Mapped I/O (MMIO):** I/O device registers are mapped into the regular memory

address space. CPU uses standard load/store instructions to communicate. Used in most modern architectures (ARM).

- **Port-Mapped I/O (PMIO):** Separate I/O address space. Requires special instructions (IN/OUT in x86). Legacy x86 feature.

MMIO advantages: Uses same cache and protection mechanisms as memory; no special instructions needed.

Q52. What is a DMA Controller and how does it work?

A:

1. CPU programs DMA controller: source address, destination address, byte count, transfer type.
2. CPU continues executing instructions.
3. DMA controller takes over the system bus in cycles (cycle stealing or burst mode).
4. Data transferred directly: Device ↔ Memory.
5. DMA controller interrupts CPU when transfer is complete.
6. CPU processes the completed I/O result.

12. Parallel Computing

Q53. What is Flynn's Taxonomy?

A: A classification of computer architectures based on instruction and data streams:

Class	Instruction	Data	Description	Example
SISD	Single	Single	Traditional sequential CPU	Classic uniprocessor
SIMD	Single	Multiple	Same op on many data	GPU, SSE/AVX instructions
MISD	Multiple	Single	Rare (fault tolerance)	Some pipeline designs
MIMD	Multiple	Multiple	Multiple CPUs, each executing independently	Multicore CPUs, clusters

Q54. What is the difference between multicore and multiprocessor systems?

A:

- **Multicore:** Multiple CPU cores on a **single chip**. Shared L3 cache and memory controller. Low communication latency. Common in laptops, servers.
 - **Multiprocessor (SMP — Symmetric Multiprocessing):** Multiple separate CPU chips sharing the **same memory bus**. Higher latency between sockets. Used in high-end servers.
 - **NUMA (Non-Uniform Memory Access):** Each processor has its own local memory; accessing remote memory is slower. Common in multi-socket servers.
-

Q55. What is cache coherence?

A: In multicore systems, each core has its own L1/L2 cache. If two cores cache the same memory location and one writes to it, the other core's cache becomes **stale (invalid)**.

Cache coherence protocols ensure all caches have a consistent view:

MESI Protocol (common):

- **M (Modified):** Only this cache has the data; it's dirty (different from memory).
- **E (Exclusive):** Only this cache has the data; it's clean.
- **S (Shared):** Multiple caches have a clean copy.
- **I (Invalid):** The cache line is invalid.

When a core writes: it sends an **invalidation** or **update** to other caches via the interconnect.

PART B — Computer Networking

13. Networking Fundamentals

Q56. What is a computer network?

A: A computer network is a set of interconnected devices (computers, servers, switches, routers) that can communicate and share resources (data, printers, internet access) using standard protocols.

Types by scale:

- **PAN** (Personal Area Network): Bluetooth, ~10 m.
- **LAN** (Local Area Network): Office/home, up to ~1 km.
- **MAN** (Metropolitan Area Network): City-scale, ~100 km.
- **WAN** (Wide Area Network): Global, Internet.

Types by topology: Bus, Star, Ring, Mesh, Hybrid.

Q57. What are network topologies?

A:

Topology	Description	Pros	Cons
Bus	Single shared cable	Simple, cheap	Single point of failure, collisions
Star	All devices connect to central hub/switch	Easy to add devices, fault isolation	Hub/switch is single point of failure
Ring	Each device connects to two neighbors in a circle	Predictable performance	Single break can fail entire network
Mesh	Every device connected to every other	Highly redundant	Expensive, complex wiring
Hybrid	Combination of topologies	Flexible	Complex

Most enterprise LANs use **Star** topology with switches.

Q58. What is the difference between a hub, switch, and router?

A:

Device	OSI Layer	Function
Hub	Layer 1 (Physical)	Broadcasts all frames to all ports. Dumb device. Creates one collision domain.
Switch	Layer 2 (Data Link)	Forwards frames based on MAC address table. Each port = separate collision domain. Intelligent.
Router	Layer 3 (Network)	Routes packets between different networks using IP addresses. Connects LANs to WANs.
Bridge	Layer 2	Connects two LAN segments; filters by MAC.
Gateway	Layer 7	Protocol translation between different network architectures.

Q59. What is bandwidth vs throughput vs latency?

A:

- **Bandwidth:** Maximum theoretical data transfer rate of a link. Measured in bps (Mbps,

Gbps). Capacity of the pipe.

- **Throughput:** Actual data transfer rate achieved in practice. Always $\leq$ bandwidth due to overhead, congestion, errors.
 - **Latency (Delay):** Time for a packet to travel from source to destination. Measured in ms.
 - Components: propagation delay + transmission delay + queuing delay + processing delay.
 - **Jitter:** Variation in latency. Critical for real-time applications (VoIP, video).
-

14. OSI Model

Q60. What is the OSI Model?

A: The OSI (Open Systems Interconnection) model is a conceptual framework defining how network communication should work in **7 layers**, each with specific responsibilities.

Layer 7: Application	– End-user protocols (HTTP, FTP, SMTP, DNS)
Layer 6: Presentation	– Encoding, encryption, compression (SSL/TLS, JPEG)
Layer 5: Session	– Session management, synchronization (RPC, NetBIOS)
Layer 4: Transport	– End-to-end communication, flow control (TCP, UDP)
Layer 3: Network	– Logical addressing, routing (IP, ICMP, OSPF)
Layer 2: Data Link	– MAC addressing, framing, error detection (Ethernet, Wi-Fi)
Layer 1: Physical	– Bits on wire, electrical/optical signals (Cable, NIC)

Mnemonic: "**All People Seem To Need Data Processing**" (top to bottom) or "**Please Do Not Throw Sausage Pizza Away**" (bottom to top).

Q61. Explain each OSI layer in detail.

A:

Layer 1 — Physical:

- Transmits raw bits over a physical medium.
- Defines electrical, optical, radio signals; cable types; pin layouts; voltage levels.
- Devices: NIC (hardware), repeater, hub.
- Examples: Ethernet (physical specifications), USB, Bluetooth radio.

Layer 2 — Data Link:

- Framing: wraps packets in frames with MAC addresses.
- Error detection: CRC/checksum.
- Flow control: prevents fast sender from overwhelming slow receiver.
- Media Access Control: who gets to use the shared medium (CSMA/CD, CSMA/CA).

- Sublayers: LLC (Logical Link Control) + MAC (Media Access Control).
- Devices: Switch, Bridge, NIC (firmware).
- Protocols: Ethernet, Wi-Fi (IEEE 802.11), PPP, ARP.

Layer 3 — Network:

- Logical addressing (IP addresses).
- Routing: choosing path from source to destination across multiple networks.
- Fragmentation and reassembly.
- Devices: Router, Layer-3 Switch.
- Protocols: IPv4, IPv6, ICMP, OSPF, BGP, RIP.

Layer 4 — Transport:

- End-to-end communication between processes (port numbers).
- Reliable delivery (TCP) or best-effort (UDP).
- Segmentation and reassembly.
- Flow control and congestion control.
- Protocols: TCP, UDP, SCTP.

Layer 5 — Session:

- Establishes, manages, and terminates sessions between applications.
- Synchronization checkpoints for long transfers.
- Protocols: NFS, RPC, SQL sessions, NetBIOS.

Layer 6 — Presentation:

- Data translation, encryption/decryption, compression.
- Converts between different data formats (ASCII ↔ EBCDIC).
- Protocols: SSL/TLS (encryption), JPEG, MPEG, ASCII, Unicode.

Layer 7 — Application:

- Interface between network and end-user applications.
- Provides services directly to software.
- Protocols: HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP, Telnet, SSH, SNMP.

Q62. What is PDU (Protocol Data Unit) at each layer?

A:

Layer	PDU Name
7 Application	Data / Message

Layer	PDU Name
6 Presentation	Data
5 Session	Data
4 Transport	Segment (TCP) / Datagram (UDP)
3 Network	Packet
2 Data Link	Frame
1 Physical	Bits

Each layer adds its own header (and sometimes trailer) — called **encapsulation**. At the receiver, headers are stripped layer by layer — **decapsulation**.

15. TCP/IP Model

Q63. What is the TCP/IP Model? How does it compare to OSI?

A: The TCP/IP (Internet) model is the practical model the Internet uses, with 4 layers:

TCP/IP Layer	OSI Equivalent
Application	→ Application + Presentation + Session (5, 6, 7)
Transport	→ Transport (4)
Internet	→ Network (3)
Network Access	→ Data Link + Physical (1, 2)

TCP/IP is the **actual implementation**; OSI is the **reference model**. TCP/IP is what the Internet runs on.

Q64. Why does TCP/IP have 4 layers while OSI has 7?

A: TCP/IP merges layers for simplicity:

- OSI layers 5, 6, 7 (Session, Presentation, Application) are merged into one Application layer in TCP/IP because in practice, most applications handle these concerns internally (e.g., TLS handles presentation-level encryption, application protocols manage sessions).
- OSI layers 1 and 2 (Physical, Data Link) are merged into Network Access (Link) layer because they both deal with physical communication on a single link.

16. Physical & Data Link Layer

Q65. What transmission media types exist?

A:

Guided (Wired):

- **Twisted Pair (UTP/STP):** Most common for LANs. Cat5e (1 Gbps), Cat6 (10 Gbps), Cat8 (40 Gbps).
- **Coaxial Cable:** TV cable, older Ethernet (10BASE2). Better shielding than twisted pair.
- **Fiber Optic:** Transmits light pulses. Very high bandwidth, immune to EMI, long distances.
 - **Single-mode:** Long distances (up to 100 km). One light path.
 - **Multi-mode:** Shorter distances (< 2 km). Multiple light paths.

Unguided (Wireless):

- Radio waves, microwaves, infrared.
 - Wi-Fi (2.4/5/6 GHz), Bluetooth, satellite, cellular.
-

Q66. What is CSMA/CD and CSMA/CA?

A:

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) — used in wired Ethernet (now mostly obsolete with full-duplex switches):

1. Listen before transmitting (Carrier Sense).
2. If medium idle, transmit.
3. If collision detected, stop, send jam signal.
4. Wait random backoff time (exponential backoff), retry.

CSMA/CA (Collision Avoidance) — used in Wi-Fi (IEEE 802.11):

- Can't detect collisions in wireless (hidden node problem), so tries to avoid them.
1. Sense medium; if idle, wait DIFS (Distributed Inter-Frame Space).
 2. Start random backoff counter.
 3. If medium stays idle until counter reaches zero, transmit.
 4. Wait for ACK. If no ACK, assume collision, retry with larger backoff.
-

Q67. What is a MAC address?

A: MAC (Media Access Control) address is a **48-bit hardware address** burned into every network interface card (NIC), used for communication within a local network (Layer 2).

Format: AA : BB : CC : DD : EE : FF (6 bytes in hex).

- First 3 bytes (OUI): Organizationally Unique Identifier — identifies the manufacturer.
- Last 3 bytes: Device-specific serial.

Properties:

- Globally unique (in theory).
- Operates at Layer 2 (doesn't cross router boundaries).
- Can be changed in software (MAC spoofing).

Q68. What is ARP (Address Resolution Protocol)?

A: ARP maps a **known IP address to an unknown MAC address** within a local network.

Process:

1. Host A wants to send a packet to IP 192.168.1.5 but doesn't know its MAC.
2. A broadcasts: "Who has 192.168.1.5? Tell 192.168.1.1" (ARP Request).
3. Host B (192.168.1.5) replies: "192.168.1.5 is at AA:BB:CC:DD:EE:FF" (ARP Reply, unicast).
4. A caches the result in its ARP table.

ARP Spoofing/Poisoning: Attacker sends fake ARP replies, mapping attacker's MAC to another host's IP — enables man-in-the-middle attacks.

Q69. What is the Ethernet frame structure?

A:

Preamble (7B)	SFD (1B)	Dest MAC (6B)	Src MAC (6B)	EtherType/Length (2B)	Payload (46–1500B)	CRC/FCS (4B)
---------------	----------	---------------	--------------	-----------------------	--------------------	--------------

- **Preamble:** Synchronization pattern (alternating 1s and 0s).
- **SFD:** Start Frame Delimiter — marks start of frame.
- **Dest/Src MAC:** 48-bit addresses.
- **EtherType:** Identifies protocol in payload (0x0800 = IPv4, 0x86DD = IPv6, 0x0806 = ARP).
- **Payload:** 46–1500 bytes of data (MTU = 1500 bytes).
- **CRC/FCS:** 32-bit CRC for error detection.

17. Network Layer & IP Addressing

Q70. What is an IP address?

A: An IP (Internet Protocol) address is a **logical address** assigned to a network interface for identification and location addressing.

IPv4: 32-bit, written as 4 dotted-decimal octets: 192 . 168 . 1 . 100. ~4.3 billion addresses.

IPv6: 128-bit, written as 8 groups of 4 hex digits:

2001:0db8:85a3:0000:0000:8a2e:0370:7334. Provides $\sim 3.4 \times 10^{38}$ addresses.

IP addresses have two parts:

- **Network portion:** Identifies the network.
 - **Host portion:** Identifies the specific device.
-

Q71. What is subnetting and CIDR?

A: Subnetting divides a larger network into smaller subnetworks (subnets) by borrowing bits from the host portion.

CIDR (Classless Inter-Domain Routing): Replaced the old classful addressing. Uses a **prefix length** to specify the network mask.

192.168.1.0/24:

- /24 means first 24 bits = network, last 8 bits = hosts.
- Subnet mask: 255.255.255.0
- Network: 192.168.1.0
- Broadcast: 192.168.1.255
- Usable hosts: $192.168.1.1 - 192.168.1.254 = 254$ hosts

Subnetting example: Split 192.168.1.0/24 into 4 subnets → use /26 (borrow 2 bits, $2^2=4$ subnets, each with $2^{6-2} = 62$ hosts).

Q72. What are private IP address ranges?

A: RFC 1918 defines private address ranges not routable on the public Internet:

Range	CIDR	Addresses
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16,777,216
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536

Also: 127.0.0.0/8 = loopback (localhost = 127.0.0.1), 169.254.0.0/16 = APIPA (link-local, auto-assigned when DHCP fails).

Q73. What are the classes of IPv4 addresses (classful)?

A:

Class	Range	Default Mask	Purpose
A	1.0.0.0 – 126.255.255.255	/8	Large networks (126 networks, 16M hosts each)
B	128.0.0.0 – 191.255.255.255	/16	Medium networks (16K networks, 65K hosts each)
C	192.0.0.0 – 223.255.255.255	/24	Small networks (2M networks, 254 hosts each)
D	224.0.0.0 – 239.255.255.255	—	Multicast
E	240.0.0.0 – 255.255.255.255	—	Reserved/Experimental

Classful addressing is mostly replaced by CIDR today.

Q74. What is IPv6 and why is it needed?

A: IPv4 exhaustion (only ~4.3 billion addresses — insufficient for billions of devices) drove IPv6 adoption.

IPv6 improvements:

- **128-bit address space:** 3.4×10^{38} addresses.
 - **No NAT required:** Every device can have a globally unique address.
 - **Simplified header:** Fixed 40-byte header (IPv4 header is variable). Faster routing.
 - **Built-in IPSec:** Security is mandatory, not optional.
 - **Stateless Address Autoconfiguration (SLAAC):** Devices can auto-configure without DHCP.
 - **No broadcast:** Uses multicast and anycast instead.
 - **Better QoS:** Flow label field.
-

18. Routing & Switching

Q75. What is routing? Types of routing?

A: Routing is the process of **selecting the best path** for packets to travel from source to destination across multiple networks.

Static Routing: Manually configured routes. Simple, no overhead, but doesn't adapt to failures.

Dynamic Routing: Routers automatically discover and adapt routes using routing protocols.

- **Distance-Vector protocols:** Each router knows only its neighbors' distances. Shares routing tables. Slow convergence.
 - Examples: **RIP** (Routing Information Protocol) — max 15 hops, updates every 30s.
 - **Link-State protocols:** Each router builds a complete topology map (knows the entire network). Faster convergence.
 - Examples: **OSPF** (Open Shortest Path First) — uses Dijkstra's algorithm. Industry standard IGP.
 - **Path-Vector protocols:** Like distance-vector but stores entire path to prevent loops.
 - Examples: **BGP** (Border Gateway Protocol) — used for inter-AS (Internet) routing.
-

Q76. What is the difference between IGP and EGP?

A:

- **IGP (Interior Gateway Protocol):** Used for routing within a single Autonomous System (AS — a network under one administrative domain). Examples: OSPF, EIGRP, RIP, IS-IS.
- **EGP (Exterior Gateway Protocol):** Used for routing between different Autonomous Systems. The Internet's backbone. Example: **BGP-4** — the only EGP in widespread use.

An ISP's internal network uses OSPF; the ISP connects to other ISPs using BGP.

Q77. What is the difference between routing and switching?

A:

- **Switching** (Layer 2): Forwards frames within the same network based on **MAC addresses**. Uses a MAC address table (CAM table). Fast hardware-based forwarding.
- **Routing** (Layer 3): Forwards packets between different networks based on **IP addresses**. Uses a routing table. Involves software or specialized hardware (ASICs).

Modern Layer-3 switches combine both: wire-speed routing using ASICs.

Q78. What is the difference between a collision domain and a broadcast domain?

A:

- **Collision Domain:** A network segment where simultaneous transmissions cause collisions. Each switch port = separate collision domain (switches isolate collisions). Hubs = one big collision domain.
 - **Broadcast Domain:** A set of devices that receive each other's Layer-2 broadcast frames. Routers separate broadcast domains. Switches do NOT separate broadcast domains by default (VLANs do).
-

Q79. What is VLAN (Virtual LAN)?

A: A VLAN logically segments a physical switch into multiple virtual switches, separating broadcast domains without requiring separate hardware.

Benefits:

- **Security:** Isolate sensitive departments (finance, HR).
- **Performance:** Reduce broadcast traffic.
- **Flexibility:** Users in different physical locations can be in the same VLAN.

VLAN Trunking (IEEE 802.1Q): Tags frames with a 12-bit VLAN ID to carry multiple VLANs over one physical link between switches.

Inter-VLAN routing requires a router or Layer-3 switch.

Q80. What is Spanning Tree Protocol (STP)?

A: STP (IEEE 802.1D) prevents **Layer-2 loops** in networks with redundant paths between switches. Without STP, frames would loop forever, causing broadcast storms and network collapse.

STP elects a **Root Bridge** (lowest Bridge ID = Priority + MAC). Each non-root switch finds the shortest path to the root. Redundant ports are placed in **Blocking** state.

Port states: Blocking → Listening → Learning → Forwarding → Disabled.

Converges in ~30–50 seconds (slow). **RSTP (802.1w)** converges in < 1 second. **MSTP (802.1s)** maps VLANs to STP instances.

19. Transport Layer — TCP & UDP

Q81. What is the difference between TCP and UDP?

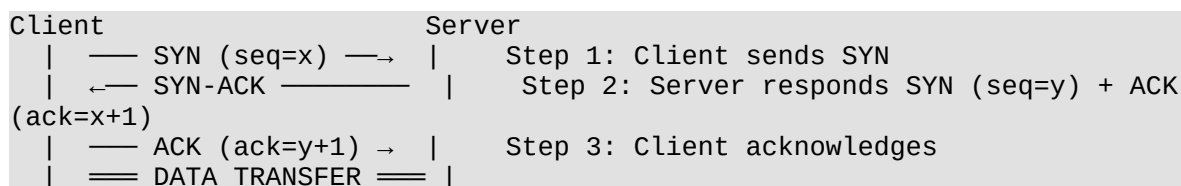
A:

Feature	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery, acknowledgments, retransmission	Best-effort, no guarantee
Ordering	In-order delivery guaranteed	No ordering
Flow Control	Yes (sliding window)	No
Congestion Control	Yes	No
Speed	Slower (overhead)	Faster

Feature	TCP	UDP
Header size	20–60 bytes	8 bytes
Use cases	HTTP, FTP, SSH, Email	DNS, VoIP, video streaming, gaming

Q82. Explain the TCP 3-way handshake.

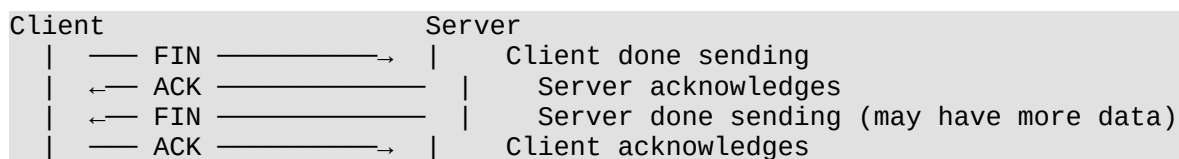
A: TCP establishes a connection before data transfer:



Purpose: Synchronize sequence numbers, negotiate parameters, establish connection state at both ends.

Q83. How does TCP terminate a connection (4-way handshake)?

A: TCP teardown requires 4 steps because each direction closes independently:



Client enters **TIME_WAIT** state (lasts $2 \times \text{MSL} = \sim 60\text{--}120$ sec) to ensure the final ACK is received and old duplicate packets are gone before the port is reused.

Q84. What is TCP flow control?

A: Flow control prevents a fast sender from overwhelming a slow receiver.

TCP uses a **sliding window** mechanism:

- The receiver advertises a **receive window (rwnd)** in each ACK — the amount of buffer space available.
- The sender limits the amount of unacknowledged data to $\min(\text{cwnd}, \text{rwnd})$ where cwnd = congestion window.
- If rwnd = 0: sender stops, sends periodic probes (zero window probes).

Q85. What is TCP congestion control?

A: Congestion control prevents the sender from overwhelming the **network** (not just the receiver).

Four Phases:

1. **Slow Start:** Begin with $cwnd = 1 \text{ MSS}$. Double $cwnd$ every RTT (exponential growth) until $ssthresh$ is reached.
2. **Congestion Avoidance:** $cwnd > ssthresh \rightarrow$ increase $cwnd$ by 1 MSS per RTT (linear growth).
3. **Fast Retransmit:** On receiving 3 duplicate ACKs (indicating loss), retransmit without waiting for timeout.
4. **Fast Recovery:** After fast retransmit, set $ssthresh = cwnd/2$, $cwnd = ssthresh + 3$ (TCP Reno). Resume congestion avoidance.

On timeout: $ssthresh = cwnd/2$, $cwnd = 1 \text{ MSS}$, restart slow start.

Q86. What is a TCP socket? What are socket states?

A: A socket is an endpoint for network communication, identified by (**IP address, Port number, Protocol**).

Key TCP socket states:

- **LISTEN:** Server waiting for incoming connections.
 - **SYN_SENT:** Client has sent SYN.
 - **SYN_RECEIVED:** Server received SYN, sent SYN-ACK.
 - **ESTABLISHED:** Connection open, data transfer possible.
 - **FIN_WAIT_1/2:** Sent FIN, waiting for ACK/remote FIN.
 - **CLOSE_WAIT:** Received FIN, waiting for local app to close.
 - **TIME_WAIT:** Waiting to ensure remote ACK received.
 - **CLOSED:** No connection.
-

20. Application Layer Protocols

Q87. How does HTTP work?

A: HTTP (HyperText Transfer Protocol) is a request-response protocol for transferring web content.

HTTP Request:

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
```

HTTP Response:

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1234

<html>...</html>
```

Methods: GET (read), POST (create/submit), PUT (replace), PATCH (partial update), DELETE, HEAD (headers only), OPTIONS.

Status Codes:

- 1xx: Informational
- 2xx: Success (200 OK, 201 Created, 204 No Content)
- 3xx: Redirection (301 Moved Permanently, 302 Found)
- 4xx: Client Error (400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found)
- 5xx: Server Error (500 Internal Server Error, 503 Service Unavailable)

Q88. What is the difference between HTTP/1.1, HTTP/2, and HTTP/3?

A:

Feature	HTTP/1.1	HTTP/2	HTTP/3
Transport	TCP	TCP	QUIC (UDP-based)
Multiplexing	No (one request per connection; pipelining fragile)	Yes (multiple streams in one TCP connection)	Yes
Head-of-line blocking	Yes	TCP-level HOL blocking	No (stream-level isolation)
Header compression	No	HPACK compression	QPACK compression
Server Push	No	Yes	Yes
Connection setup	1–3 RTT	1–2 RTT	0–1 RTT

HTTP/3 uses **QUIC** (built on UDP) to eliminate TCP's head-of-line blocking and reduce connection setup latency.

Q89. What is HTTPS and how does TLS work?

A: HTTPS = HTTP over TLS (Transport Layer Security). Provides encryption, authentication, and integrity.

TLS Handshake (TLS 1.3, simplified):

1. Client sends **ClientHello**: TLS version, cipher suites, random nonce, key share.
2. Server sends **ServerHello**: Chosen cipher suite, random nonce, key share, certificate.
3. Client verifies server's **certificate** against trusted CAs (Certificate Authorities).

4. Both derive **session keys** from key shares (ECDHE — no secret is sent over the wire).
5. Both send **Finished** messages (encrypted with session key).
6. Encrypted **application data** flows.

TLS 1.3 achieves 1-RTT handshake (0-RTT for resumed sessions).

Q90. How does FTP work? What are active vs passive modes?

A: FTP (File Transfer Protocol) uses **two TCP connections**:

- **Control connection** (port 21): Send commands and responses.
- **Data connection** (port 20 or dynamic): Transfer actual file data.

Active Mode: Server initiates the data connection back to the client. Problematic with client-side firewalls/NAT (incoming connection blocked).

Passive Mode (PASV): Client initiates both connections. Server opens a random port, tells client to connect to it. Firewall-friendly.

SFTP (SSH File Transfer Protocol) and FTPS (FTP over TLS) are secure alternatives.

Q91. How does email work (SMTP, POP3, IMAP)?

A:

SMTP (Simple Mail Transfer Protocol) — port 25/587/465: Used to **send** email from client to server, and between mail servers.

POP3 (Post Office Protocol v3) — port 110/995: Downloads email to local client, typically deletes from server. Offline access. Simple.

IMAP (Internet Message Access Protocol) — port 143/993: Keeps email on server; client syncs. Supports multiple devices, folders, and search. Modern standard.

Email flow:

Sender → SMTP → Sender's Mail Server → SMTP → Recipient's Mail Server → IMAP/POP3 → Recipient

21. DNS, DHCP & NAT

Q92. How does DNS work?

A: DNS (Domain Name System) translates human-readable domain names (google.com) to IP addresses (142.250.80.78).

DNS Resolution Process:

1. Browser checks its **local cache**.

2. OS checks its **hosts file** and **DNS cache**.
3. Query sent to **recursive resolver** (usually ISP or 8.8.8.8).
4. Resolver checks its cache; if miss, queries **root nameservers** (.) → told where to find .com TLD servers.
5. Resolver queries **.com TLD servers** → told authoritative nameserver for google.com.
6. Resolver queries **google.com authoritative nameserver** → gets IP.
7. Resolver caches result (respecting TTL) and returns IP to client.

DNS Record Types:

- **A:** Domain → IPv4
 - **AAAA:** Domain → IPv6
 - **CNAME:** Alias → Canonical name
 - **MX:** Mail exchange server
 - **NS:** Authoritative nameserver
 - **TXT:** Text (used for SPF, DKIM, verification)
 - **PTR:** Reverse DNS (IP → Domain)
 - **SOA:** Start of Authority
-

Q93. What is DHCP and how does it work?

A: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration to devices.

DORA Process:

1. **Discover:** Client broadcasts "I need an IP address!" (DHCPDISCOVER).
2. **Offer:** Server broadcasts "I offer you 192.168.1.50" (DHCPOFFER).
3. **Request:** Client broadcasts "I accept 192.168.1.50" (DHCPREQUEST).
4. **Acknowledge:** Server confirms "192.168.1.50 is yours for X seconds" (DHCPACK).

DHCP provides: IP address, subnet mask, default gateway, DNS server addresses, lease duration.

DHCP uses UDP (port 67 server, 68 client). Broadcasts because client has no IP yet.

Q94. What is NAT (Network Address Translation)?

A: NAT translates private IP addresses to public IP addresses, allowing multiple devices to share a single public IP.

How it works (PAT — Port Address Translation, most common):

1. Device (192.168.1.5:3000) sends packet to 8.8.8.8:53.

2. Router replaces source with public IP + unique port: (203.0.113.1:40001).
3. Router maintains NAT table: {203.0.113.1:40001 ↔ 192.168.1.5:3000}.
4. Reply arrives at 203.0.113.1:40001.
5. Router looks up NAT table, translates back to 192.168.1.5:3000.

Benefits: Conserves IPv4 addresses, provides implicit security (unsolicited incoming connections are dropped). **Drawbacks:** Breaks end-to-end connectivity, complicates VoIP/gaming/VPN, adds latency.

22. Network Security

Q95. What is a firewall?

A: A firewall monitors and controls incoming/outgoing network traffic based on security rules.

Types:

- **Packet Filter:** Inspects IP/TCP/UDP headers only. Stateless. Fast but limited.
 - **Stateful Firewall:** Tracks TCP connection state. Only allows packets that are part of established connections. More secure.
 - **Application Layer Gateway (Proxy Firewall):** Understands application protocols (HTTP, FTP). Deep inspection but slower.
 - **Next-Generation Firewall (NGFW):** Combines stateful inspection + application awareness + IPS + TLS inspection + user identity.
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Q96. What are common network attacks?

A:

Denial of Service (DoS/DDoS):

- Floods target with traffic to exhaust resources.
- **SYN Flood:** Sends thousands of SYN packets, never completing handshakes, exhausting connection table.
- Mitigation: SYN cookies, rate limiting, scrubbing centers.

Man-in-the-Middle (MITM):

- Attacker intercepts communication between two parties.
- Methods: ARP spoofing, DNS poisoning, SSL stripping.
- Mitigation: TLS/HTTPS with certificate pinning, HSTS.

DNS Spoofing/Cache Poisoning:

- Attacker injects false DNS records into a resolver's cache.
- Mitigation: DNSSEC (cryptographically signs DNS records).

Port Scanning: Reconnaissance — identifying open ports and services. Tool: nmap.

Packet Sniffing: Capturing network traffic. Tool: Wireshark. Mitigation: Encryption.

SQL Injection / XSS: Application-layer attacks.

Replay Attack: Attacker captures and retransmits a valid authentication message. Mitigation: Nonces, timestamps.

Q97. What is a VPN (Virtual Private Network)?

A: A VPN creates an **encrypted tunnel** over a public network (Internet), allowing remote users or sites to communicate as if on a private network.

Types:

- **Remote Access VPN:** Individual user → corporate network (e.g., employees working from home). Uses SSL/TLS VPN or IPsec.
- **Site-to-Site VPN:** Connects two office networks over the Internet using IPsec tunnels.

Protocols:

- **IPsec:** Industry standard, operates at Layer 3. Two modes: Transport (encrypts payload only) and Tunnel (encrypts entire packet, adds new IP header).
 - **OpenVPN:** Open source, uses TLS. Flexible, firewall-friendly.
 - **WireGuard:** Modern, fast, minimal codebase (~4000 lines). Uses state-of-the-art cryptography.
 - **PPTP:** Legacy, broken encryption — avoid.
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Q98. What is PKI (Public Key Infrastructure)?

A: PKI is a system for creating, managing, and distributing **digital certificates** that bind public keys to identities.

Components:

- **CA (Certificate Authority):** Trusted entity that signs certificates. Root CA and Intermediate CA.
- **Certificate:** Contains public key + identity + CA signature + validity period. X.509 format.
- **CSR (Certificate Signing Request):** Request sent to CA to obtain a certificate.
- **CRL (Certificate Revocation List) / OCSP:** Mechanisms to check if a certificate has been revoked.

Used in: HTTPS, email signing, code signing, VPN authentication.

Q99. What are symmetric and asymmetric encryption in networking?

A:

Aspect	Symmetric	Asymmetric
Keys	Same key for encrypt/decrypt	Public key (encrypt) + Private key (decrypt)
Speed	Fast (AES: ~1 GB/s)	Slow (RSA: ~1 MB/s)
Key distribution	Challenge (must share key securely)	Easy (public key can be shared openly)
Examples	AES, 3DES, ChaCha20	RSA, ECC, Diffie-Hellman
Use case	Bulk data encryption	Key exchange, digital signatures, authentication

In practice (TLS): Asymmetric (ECDHE) is used for **key exchange** to securely establish a shared secret. That secret is then used to derive symmetric keys (AES) for **bulk data encryption**. Best of both worlds.

23. Wireless Networking

Q100. How does Wi-Fi work (IEEE 802.11)?

A: Wi-Fi uses **CSMA/CA** for medium access. Key parameters:

IEEE 802.11 Standards:

Standard	Band	Max Speed
802.11a	5 GHz	54 Mbps
802.11b	2.4 GHz	11 Mbps
802.11g	2.4 GHz	54 Mbps
802.11n (Wi-Fi 4)	2.4/5 GHz	600 Mbps
802.11ac (Wi-Fi 5)	5 GHz	3.5 Gbps
802.11ax (Wi-Fi 6)	2.4/5/6 GHz	9.6 Gbps
802.11be (Wi-Fi 7)	2.4/5/6 GHz	46 Gbps

Channels: 2.4 GHz has 11 channels (3 non-overlapping: 1, 6, 11); 5 GHz has many more non-overlapping channels.

Security: WEP (broken) → WPA → WPA2 (AES-CCMP, current standard) → WPA3 (SAE, stronger handshake).

24. Advanced Networking Concepts

Q101. What is QoS (Quality of Service)?

A: QoS is a set of techniques to manage and prioritize network traffic to ensure acceptable performance for critical applications.

Mechanisms:

- **Classification and Marking:** Identify and tag traffic (DSCP in IP header, 802.1p in Ethernet).
- **Queuing:** Priority queuing (PQ), Weighted Fair Queuing (WFQ), CBWFQ.
- **Traffic Shaping:** Smooth bursty traffic to conform to a rate.
- **Traffic Policing:** Drop/re-mark packets exceeding a rate.

Critical for: VoIP (requires < 150 ms latency, < 30 ms jitter), video conferencing, real-time gaming.

Q102. What is SDN (Software-Defined Networking)?

A: SDN decouples the **control plane** (routing decisions) from the **data plane** (packet forwarding), centralizing control in a software-based **SDN controller**.

Traditional networking: Each router/switch makes independent forwarding decisions. SDN: Central controller has a global view, programs all devices via a southbound API (OpenFlow).

Benefits: Centralized management, programmability, easier automation, faster deployment. Used in: Data centers (Google, Facebook), cloud providers.

Q103. What is CDN (Content Delivery Network)?

A: A CDN is a geographically distributed network of servers that caches and delivers content from locations closer to the end user.

Benefits:

- Reduced latency (serve from edge node near user).
- Reduced load on origin server.
- DDoS mitigation (absorb traffic).
- High availability.

Examples: Cloudflare, Akamai, AWS CloudFront, Fastly.

How: DNS-based routing directs users to the nearest edge server (anycast or GeoDNS).

Q104. What is BGP and how does Internet routing work?

A: BGP (Border Gateway Protocol) is the **routing protocol of the Internet** — it's the glue holding

the global Internet together.

The Internet is divided into **Autonomous Systems (AS)** — networks run by ISPs, companies, or universities, each with a unique **ASN (Autonomous System Number)**.

BGP exchanges **routing information between ASes**:

- Each AS advertises which IP prefixes it can reach.
- BGP uses a path vector approach — stores the full AS path to detect and prevent loops.
- **Route selection** is based on: local preference, AS path length, MED, origin, and routing policy.

iBGP: BGP within an AS. **eBGP**: BGP between ASes.

A BGP misconfiguration can accidentally **hijack** traffic for large portions of the Internet.

Q105. What is the difference between unicast, broadcast, multicast, and anycast?

A:

Type	Destination	Description	Example
Unicast	Single specific host	One-to-one communication	HTTP request to a web server
Broadcast	All hosts on subnet	One-to-all on local network	ARP request, DHCP discover
Multicast	Group of interested hosts	One-to-many (subscribers)	Live video streaming (IPTV)
Anycast	Nearest member of a group	Packet delivered to the topologically nearest node in a group	DNS root servers (same IP, different locations), CDN

IPv6 eliminates broadcast, replacing it with multicast and anycast.

☐ Quick Reference Cheat Sheets

Computer Organization

Concept	Key Point
Two's Complement	Invert bits + 1; range: $-2^{(n-1)}$ to $2^{(n-1)}-1$
IEEE 754	Sign + Biased Exponent + Mantissa
NAND/NOR	Universal gates
Von Neumann	Single shared memory for code + data
RISC vs CISC	Simple fixed instructions vs complex variable instructions

Concept	Key Point
Pipeline stages	IF → ID → EX → MEM → WB
Data Hazards	RAW, WAR, WAW; solved by forwarding/stalling
Cache mapping	Direct, Fully Associative, Set Associative
Cache replacement	LRU, FIFO, Random
Write policies	Write-through vs Write-back
Virtual memory	Page tables + TLB + page faults
DMA	Device ↔ Memory without CPU involvement
MESI	Cache coherence protocol for multicore

Networking

Concept	Key Point
OSI Layers	7: App → Pres → Sess → Trans → Net → Data → Phys
TCP/IP Layers	4: App → Transport → Internet → Network Access
Hub / Switch / Router	L1 broadcast / L2 MAC / L3 IP
ARP	IP → MAC on local network
Subnetting	Borrow host bits for network; /24 = 254 hosts
Private IPs	10.x, 172.16-31.x, 192.168.x
TCP 3-way	SYN → SYN-ACK → ACK
TCP 4-way close	FIN → ACK → FIN → ACK
Congestion control	Slow Start → Congestion Avoidance → Fast Recovery
DNS resolution	Root → TLD → Authoritative → IP
DHCP	DORA: Discover → Offer → Request → Ack
NAT/PAT	Private IP + Port → Public IP + Port
TLS	Asymmetric for key exchange → Symmetric for data
CSMA/CD	Wired Ethernet collision handling
CSMA/CA	Wi-Fi collision avoidance
BGP	Inter-AS routing protocol (Internet backbone)
VLAN	Logical LAN segmentation via 802.1Q tagging
STP/RSTP	Prevents Layer-2 loops in switched networks

Master these concepts and you'll be well-prepared for any Computer Organization & Networking interview!